Rwanda Explorer Travel App Report

1. Introduction

This project is a multi-screen Flutter Travel App focused on destinations in Rwanda. All data is hard-coded in Dart files, and all images are loaded from local assets.

2. Lab Objective Coverage

The app implements a complete travel UI with navigation flow Home -> Detail -> Booking -> Back. It emphasizes complex layouts, widget composition, reusable components, and responsive spacing.

3. Screenshots

Insert screenshots directly under each item below:

Home Screen (full page)

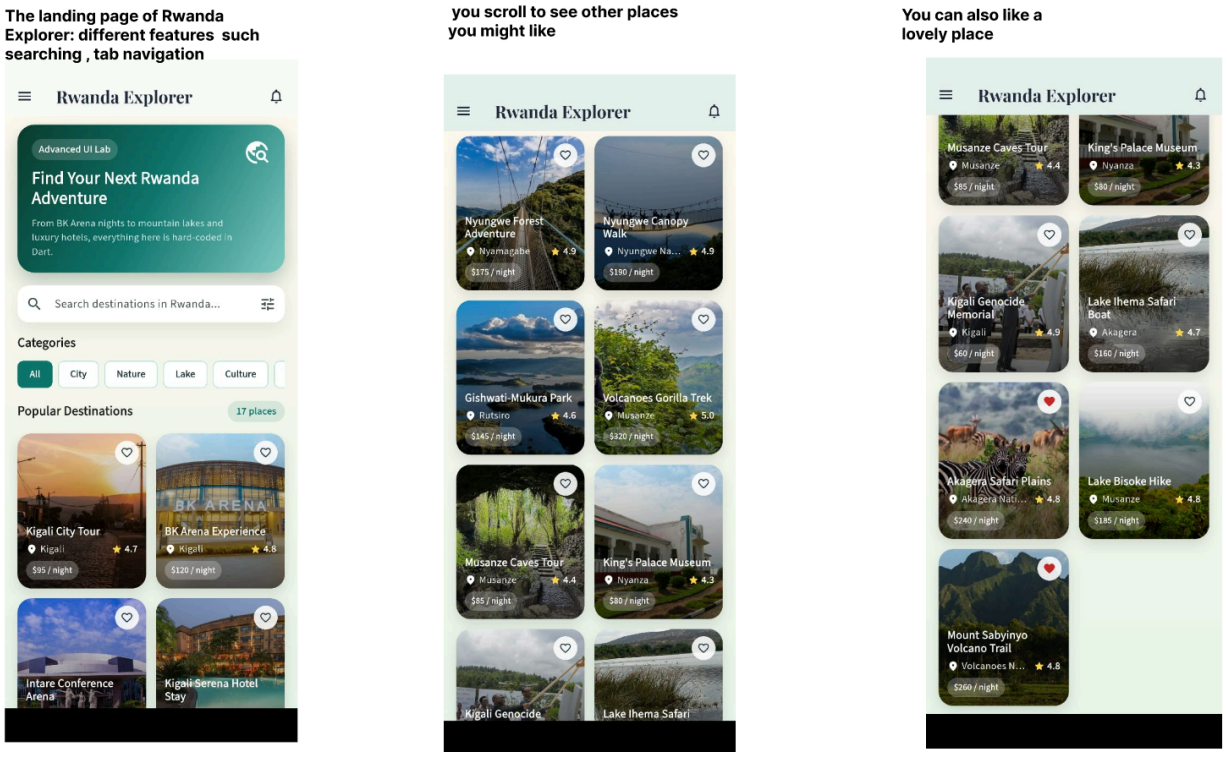

Figure 1: Home Screen (full page)

Detail Screen and booking

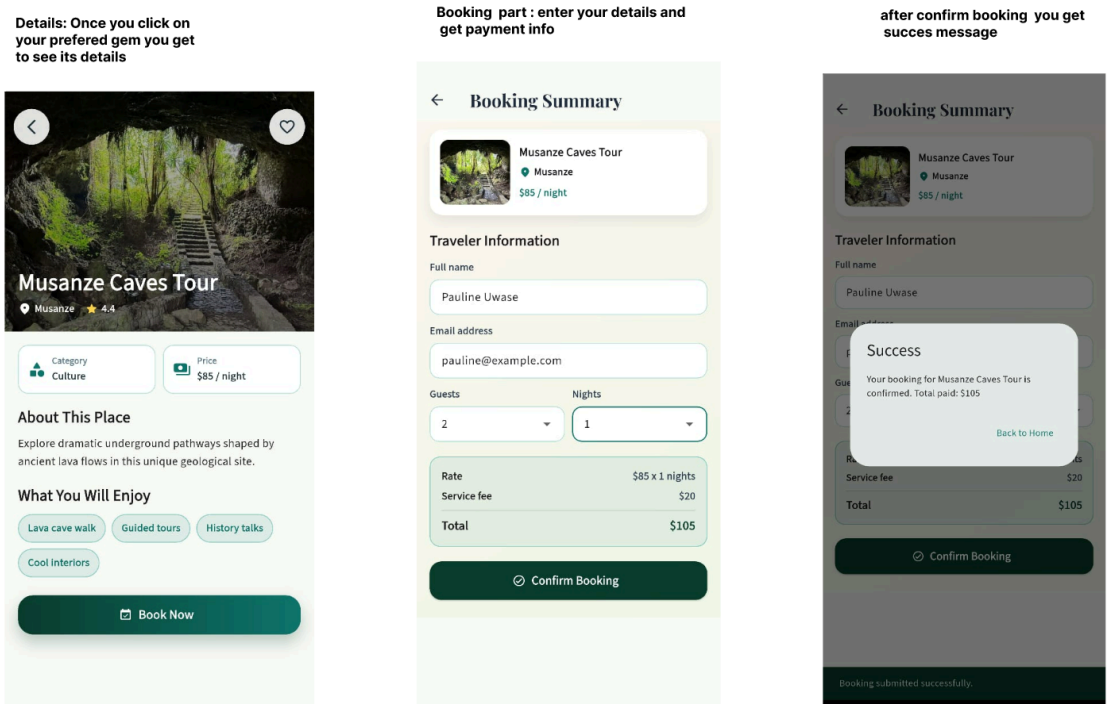

Figure 2: Detail Screen and booking

3. Why this Layout Choices

3.1 Home Screen

Dashboard-style layout was chosen to improve discovery and visual hierarchy.
The hero banner introduces app context and creates a strong first impression.
Search bar and category chips support quick filtering and better UX.
Card grid with overlays balances image-rich presentation and metadata readability.

3.2 Detail Screen

Large header image creates immersion and destination emphasis.
Gradient overlay improves text legibility on top of images.
Content sections separate key information: category, price, description, highlights.
Book Now CTA is prominent to guide users to booking flow.

3.3 Booking Screen

Top summary card keeps selected destination context visible.
Form-like structure improves data entry clarity.
Price summary block increases transparency before confirmation.
Confirmation feedback via SnackBar/Dialog validates user action.

4. Hard-Coded Data Implementation: Travel Destinations

All travel data is hard-coded in a dedicated Dart data file, including title, location, category, description, image path, price, rating, and highlights.

```
const List<String> travelCategories = <String>[
  'All',
  'City',
  'Nature',
  'Lake',
  'Culture',
  'Hotel',
  'Entertainment',
];

const List<TravelDestination> travelDestinations = <TravelDestination>[
  TravelDestination(
    id: 'kigali-city',
    title: 'Kigali City Tour',
    location: 'Kigali',
    category: 'City',
    description:
        'Explore clean boulevards, modern cafes, art spaces, and vibrant neighborhood marke
    imageAsset: 'assets/images/Kigali-City.webp',
    pricePerNight: 95,
    rating: 4.7,
    highlights: <String>['City views', 'Craft markets', 'Coffee spots', 'Night vibes'],
  ), // TravelDestination
```

Figure3: places stored in a list

5. List of Widgets Used

Every thing in flutter is a widget(every visual element and even some structural elements in this Flutter app are represented as widgets) This app uses many widgets.Main widgets include: MaterialApp, ThemeData, ColorScheme, Scaffold, AppBar, SafeArea, CustomScrollView, SliverToBoxAdapter, SliverGrid, ChoiceChip, TextField, Text, Icon, Row, Column, Container, Padding, SizedBox, Stack, Positioned, ClipRRect, Image.asset, InkWell, SingleChildScrollView, Wrap, ElevatedButton, Form, TextFormField, DropdownButtonFormField, Divider, SnackBar, AlertDialog, Navigator, MaterialPageRoute.

6. Custom Reusable Widget Classes

Reusable components were created for modularity and maintainability, including destination cards, search section, hero section, info cards, form labels, and summary rows.

7. Navigation Flow Diagram

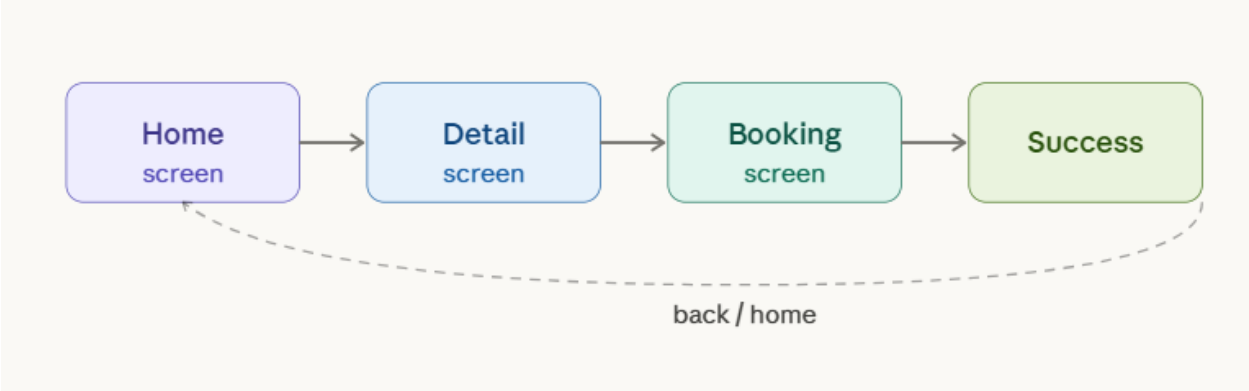

Figure 3:Navigation Flow Diagram

Conclusion

The Rwanda Explorer app allowed us to learn more about flutter widgets and how to style them. It allowed us to learn how to write clean codes with reusable widgets like cards and building responsive layouts with clear multiple screen user flow.